

Local AI Research Agent

End-to-End Setup & Deployment Guide (n8n + Flask + Claude API + ngrok + Gmail)

TECH STACK

Python Flask

Claude API

n8n

DuckDuckGo

ngrok

PROJECT GOAL

Automate local web research & email executive reports via schedule trigger

Phase 1: Local Backend & Agent Development

STEP 1 Project Directory & Environment Setup

Set up a dedicated project folder and install the required dependencies for web searching, LLM processing, and local server setup.

```
mkdir ai-research-agent
cd ai-research-agent

# Install core python packages
pip install flask duckduckgo-search anthropic requests
```

STEP 2 Build Python Research Server (`agent.py`)

Develop the local Flask application containing live search fetching, error handling/fallback, Claude API synthesis, and the HTTP POST endpoint (`/research`).

- Integrate `ddgs.text()` for live web context collection.
- Include mock search fallback to ensure execution never returns empty on rate-limit blocks.
- Expose `/research` endpoint to accept JSON payloads containing `topic` and `num_results`.

Phase 2: Local Server Execution & Public Tunneling

STEP 3 Start Local Flask Server

Navigate to your project directory and start the Flask server listening on port `5000`.

```
cd ai-research-agent
python agent.py --server
```

Important

Keep this terminal session running throughout execution. Do not close it.

STEP 4 Expose Port via ngrok Tunnel

Open a **second terminal window** and launch ngrok to create a secure HTTPS tunnel to port 5000:

```
ngrok http 5000
```

Copy the generated forwarding URL from the terminal output (e.g., `https://compare-illusive-statistic.ngrok-free.dev`).

Phase 3: n8n Workflow Construction

STEP 5 Configure Schedule Trigger & Set Topic Nodes

- **Schedule Trigger:** Set execution interval (e.g., Days/Weeks).
- **Set Topic Node:** Define the target research query (e.g., `AI Automation Trends 2026`).

STEP 6 Configure "Call Research Agent" HTTP Node

Set up the HTTP Request node to forward the topic payload to your local backend:

- **Method:** `POST`
- **URL:** `https://<your-ngrok-subdomain>.ngrok-free.dev/research`
- **Body Parameters:**
 - `topic` = `AI Automation Trends 2026`
 - `num_results` = `5`

STEP 7 Configure Gmail Node for Clean HTML Delivery

Ensure the final output renders cleanly in recipient inboxes rather than showing plain unformatted markdown glued together:

- Set **Email Type / Body Type** to `HTML`.
- Wrap the research payload output inside a formatted HTML block:

```
<div style="font-family: Arial, sans-serif; line-height: 1.6; color: #333;">
  <pre style="white-space: pre-wrap; font-family: inherit;">{{ $json.report }}</pre>
</div>
```

STEP 8 Test, Publish & Automate

1. Click **Execute workflow** in n8n to test end-to-end payload movement.
2. Verify green checkmarks across all four nodes (Schedule → Set Topic → HTTP Request → Gmail).
3. Toggle the top-right **Publish** switch in n8n to activate active background automation.